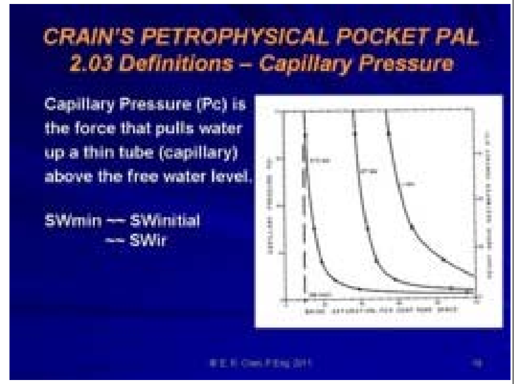
CAPILLARY PRESSURE -- Pc

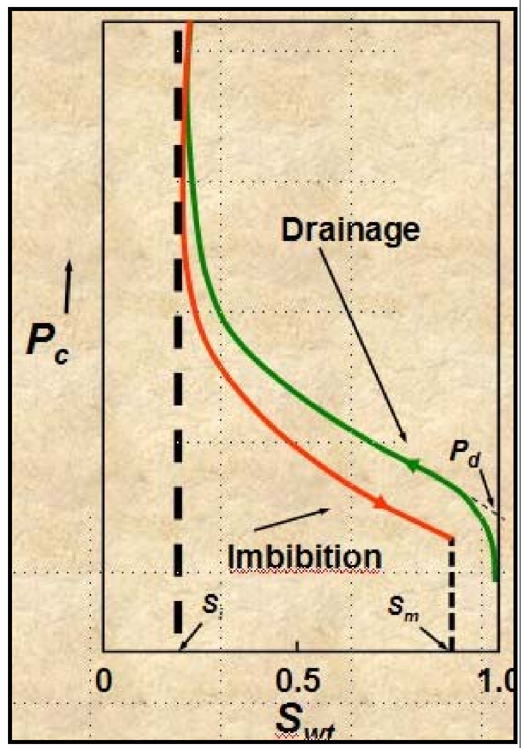
	This Page →	Cap Pres Basics	Sat-Height	Measurement	J-Function	Example
	See Also →	Water Saturation	Buckle's SW	Wettability	Courses	Site Map

CAPILLARY PRESSURE

Capillary pressure is a measurement of the force that draws a liquid up a thin tube, or capillary. Fluid saturation varies with the capillary pressure, which in turn varies with the vertical height above the free water level. Typically, laboratory measurements of capillary pressure are plotted on linear X - Y coordinate graph paper, as shown at the right.



Accumulation of hydrocarbon in a reservoir is a drainage process and production by aquifer drive or waterflood is an imbibition process. The capillary pressure curve is different for these two processes.



DRAINAGE
Fluid flow process in which the saturation of the nonwetting phase increases. Mobility of nonwetting fluid phase increases as nonwetting phase saturation increases.

IMBIBITION
Fluid flow process in which the saturation of the wetting phase increases. Mobility of wetting phase increases as wetting phase saturation increases

There are four key parameters that are related to a capillary curve :

S_i = irreducible wetting phase saturation
S_m = 1 - residual non-wetting phase saturation
P_d = displacement pressure, the pressure required to force non-wetting fluid into largest pores
LAMDA = pore size distribution index; determines shape of capillary pressure curve

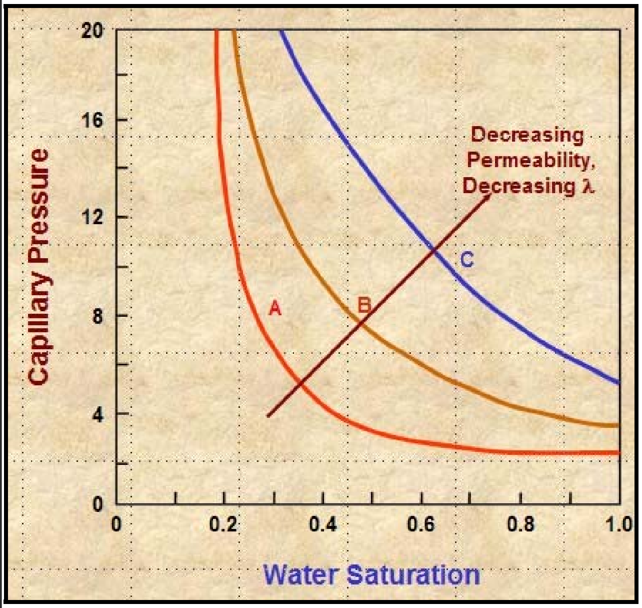
Si is the initial, or irreducible, water saturation in a hydrocarbon-bearing, water-wet, reservoir at initial pressure, prior to the start of production. It is termed SWir elsewhere in this Handbook.

(1 - Sm) is the residual oil saturation in a fully depleted water-wet reservoir, called Sor elsewhere in this Handbook.

(1 - SI) or (1 - SWir) is the non-wetting phase saturation at initial conditions prior to production. In a water-wet reservoir with a water drive, water saturation (Sw) increase above Si as production proceeds. At any time, saturation of the non-wetting phase (So) equals (1 - Sw). In a gas expansion drive reservoir Sw = (1 - So) stays relatively constant over time. Capillary pressure curves help to explain the behaviour of water drive reservoirs but do little for gas expansion or gas cap reservoirs with no water drive.

Petrophysicists use cap pressure minimum saturation (SWir) and residual oil saturation (Sor) to help

calibrate log derived water saturation in oil and gas reservoirs above the transition zone, and to help detect depleted reservoirs. It will not help calibrate SW in partially depleted zones.



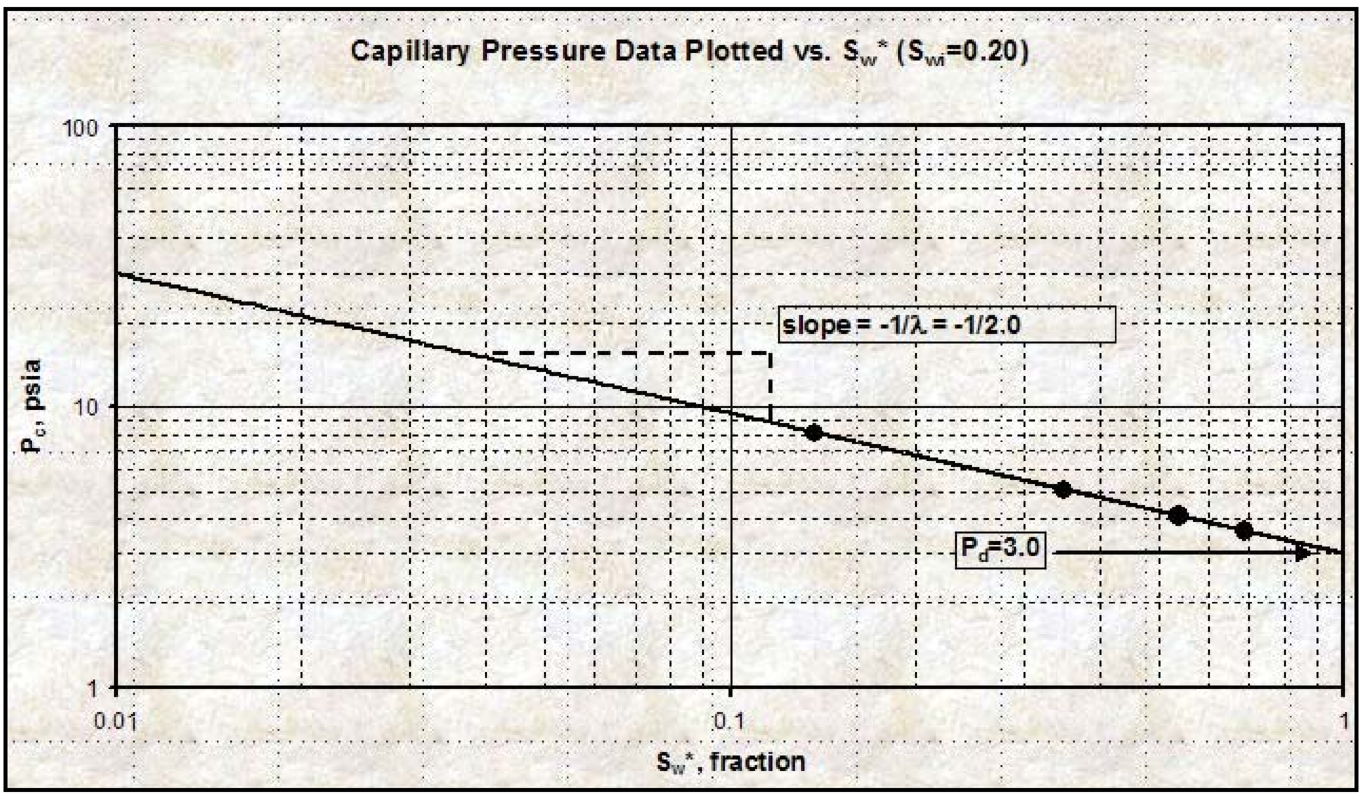
LAMDA increases with decreasing permeability, poor grain sorting, smaller grain size, and usually with lower porosity. These effects shift the cap pressure curve upward and to the right, resulting in higher SWir values.

A capillary pressure curve on Cartesian coordinates is difficult to fit with simple equations. By transforming the SW axis to Sw* and plotting Pc vs Sw* on log - log graph paper, the curves become straight lines.

1: Sw* = (Sw - SWir) / (1 - SWir - Sor)

2: Pc = Pd * (Sw*) ^ (1 / LAMDA)

The slope of the line is (1 / LAMBDA) and the intercept at 100% Sw* is Pd. SWir is obtained from the Cartesian plot. Steeper slope equals higher LAMDA equals poorer quality rock. The typical range of LAMDA is 0.5 for good quality sands to 4 or 8 for poor quality sands and carbonates. Type curve matching of Sw* can be used to assess cap pressure curves and reservoir quality.



Plot of Pc versus Sw* on log - log scale.

If a glass capillary tube is placed in a large open vessel containing water, the combination of surface tension and wettability of tube to water will cause water to rise in the tube above the water level in the container outside the tube. The water will rise in the tube until the total force acting to pull the liquid upward is balanced by the weight of the column of liquid being supported in the tube. Assuming the radius of the capillary tube is R, the total upward force Fup, which holds the liquid up, is equal to the force per unit length of surface times the total length of surface:

3: Fup = 2 * PI * R * SIGgw * cos (THETA)

The upward force is counteracted by the weight of the water, which is a downward fore equal to mass times

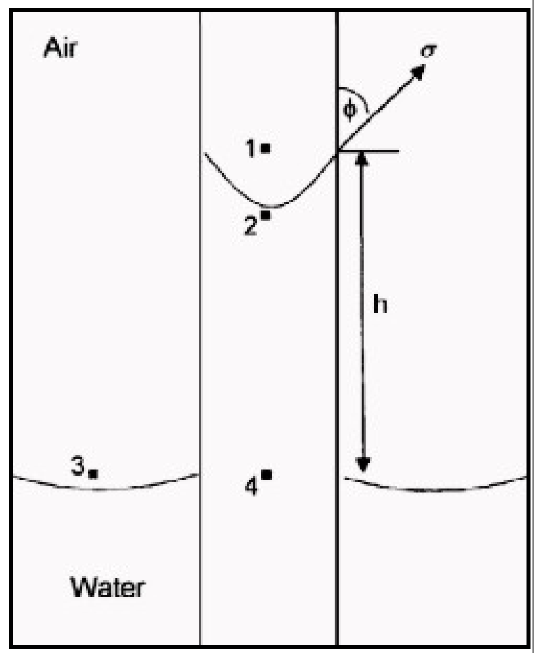
acceleration:
    4: Fdown = PI * R^2 * H * (DENSwtr - DENSair) * G

Assume density of air is negligible and set Fup = Fdown, solve for surface tension:
    5: SIGgw = (R * H * DENSwtr * G) / (2 * cos (THETA))

In the more general case of oil and water, the equation becomes:
    6: SIGow = (R * H * (DENSwtr - DENSoil) * G) / (2 * cos(THETA))

Where:
  PI = 3.1416.....
  SIGgw = = surface tension between air and water (dynes/cm)
  THETA = contact angle
  R = capillary radius (cm)
  H = height of water in capillary (cm)
  DENSwtr = density of water (gm/cc)
  DENSair = density of air (gm/cc)
  G = acceleration of gravity = 980.7 (cm/sec^2)

Rearranging equation 5:
    7: H = (SIGgw * 2 * cos (THETA)) / (R * DENSwtr * G)



The pressure difference across the interface between Points 1 and 2 is the capillary pressure:
    8: P2 = P4 – G * H * DENSwtr
    9: P1 = P3 – G * H * DENSair
    10: Pc = P1 – P2

The pressure at Point 4 within the capillary tube is the same as that at Point 3 outside the tube.
    11: Pc = (P3 – G * H * DENSair) – (P4 – G * H * DENSwtr)
    12: Pc = G * H * (DENSwtr – DENSair)
    13: Pc = G * H * ΔDENS

Where:
  Pc = capillary pressure (dynes/cm2)
  ΔDENS = density difference between the wetting and nonwetting phase.(gn/cc)

Combining equations 7 and 12:
    14: Pc = (SIGgw * 2 * cos (THETA)) / R

To convert Pc in dynes/cm2 to psi, multiply by 1.45 * 10^-5.

SATURATION - HEIGHT CURVES

To convert from laboratory (air-brine) measurements to reservoir conditions, we need to use the following relationship:
    15: Pc_res = Pc_lab * (SIGow * cos (THETAow)) / (SIGgw * cos (THETAgw))

Typical values for air-brine conversion to oil-water are:
  SIGow = 24 dynes/cm2
  THETAow = 30 deg
  SIGgw = 72 dynes/cm2

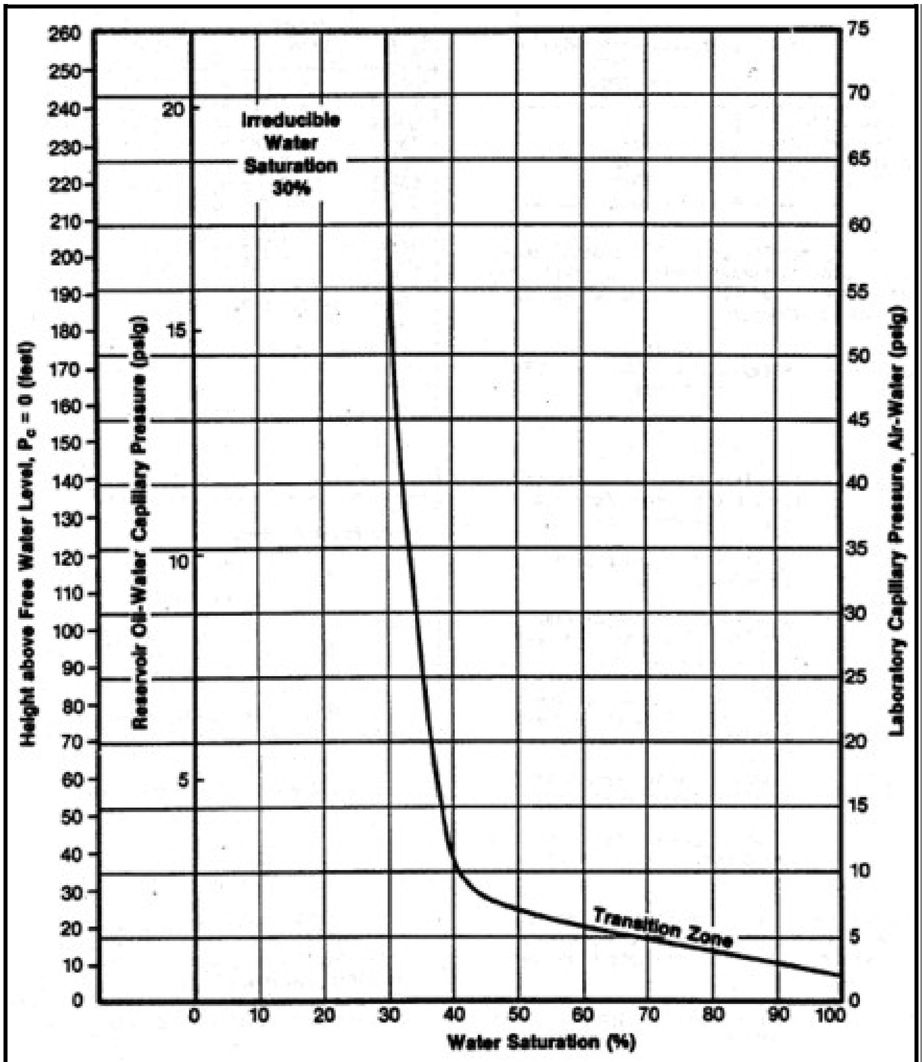
  THETAgw = 0 deg
Giving: Pc_res = 0.289 * Pc_lab

Solving equation 13 for H, and using reservoir (oil-water) Pc values:

16: H = KP15 * Pc_res / ΔDENS

Where:

Pc_res = capillary pressure at reservoir (psi or KPa)
H = capillary rise (ft or meters)
ΔDENS = density difference (gm/cc)
KP15 = 2.308 (English units)
KP15 = 0.1064 (Metric units)



<== Example of conversion of lab air-brine capillary pressure data to reservoir conditions, then into saturation-height H; results plotted in graph above..

When H is calculated at a number of points on the Pc curve, the resulting graph of H vs SW is known as a saturation-height curve and can be plotted on a depth plot of log data or results by setting H = 0 at the base of transition zone on the logs. This assumes a uniform porosity-permeability regime, which is seldom encountered in real life, so more complicated methods are needed to superimpose the saturation values from multiple Pc curves.

Sw %	Pc_lab	Pc_res	H
100	2	0.578	6.9
90	3	0.867	10.4
80	4	1.16	13.9
70	5	1.45	17.4
60	6	1.73	20.8
50	7	2.02	24.2
45	8	2.31	27.7
40	10	2.89	35
35	27	7.8	94
30	75	21.7	260

All of the above assumes the lab data is an air-brine measurement. For mercury injection capillary, pressure (MICP) measurements, the density of the non-wetting phase (mercury) is 13.5 g/cc, so ΔDENS is much larger than the air-water case. As a result, Pc values from an MICP measurement are about 12.5 times higher than an air brine measurement (for the same SW value in the same core plug). To compare an air-brine cap pressure curve to an MICP curve, it is merely necessary to change the Pc scale on one of the graphs by the appropriate factor, or to convert both Pc scales to a saturation-height scale.

MEASURING CAPILLARY PRESSURE

Capillary pressure can be measured in the laboratory in four different ways:

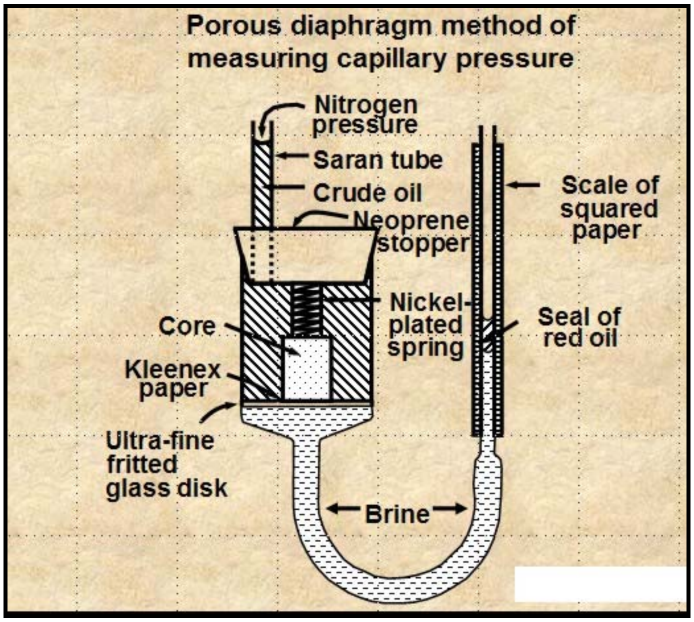
Porous diaphragm method
Mercury injection method
Centrifuge method

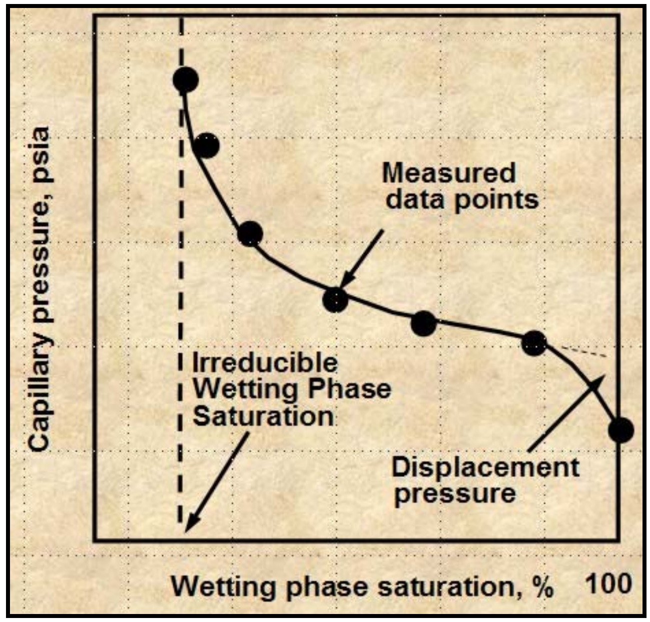
Dynamic method

Detailed operation of the laboratory equipment is beyond the scope of this Handbook. The illustrations are not fully self-explanatory, but the general principles are relatively visible.

POROUS DIAPHRAGM METHOD

The apparatus and sample data set are shown below. The method is very accurate but it can take days to months to get a complete cap pressure curve.

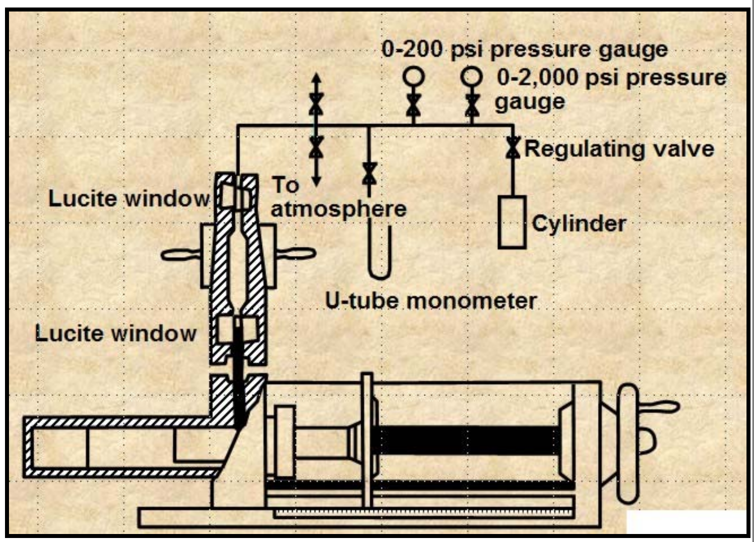




Porous Diaphragm apparatus and results

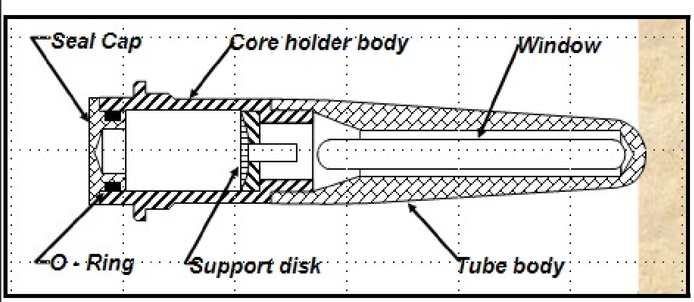
MERCURY INJECTION METHOD

The apparatus is shown at right. The method is reasonably accurate and takes minutes to hours to get a complete cap pressure curve. The core sample cannot be re-used for any purpose and require special disposal procedures due to the mercury. A conversion factor is need to get an equivalent air-brine capillary pressure, comparable to that from a porous plate method.



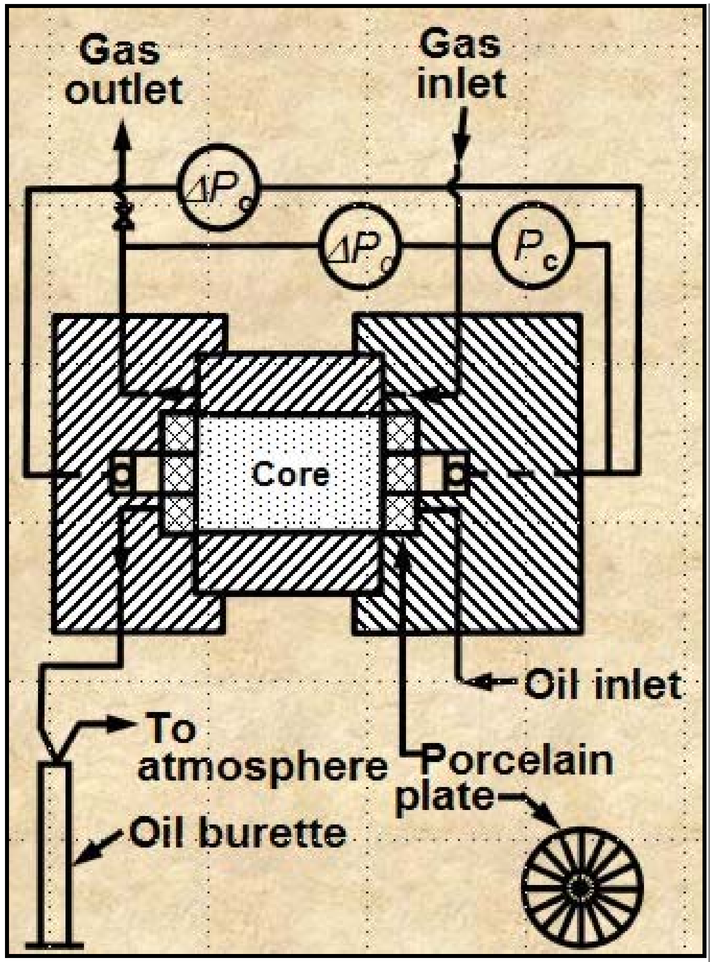
CENTRIFUGAL METHOD

The apparatus is shown below. The method is reasonably accurate and takes hours to days to get a complete cap pressure curve. Data analysis is complicated and can lead to errors.



DYNAMIC METHOD

The apparatus is shown at right. The method is reasonably accurate and simulates actual reservoir flow when whole core is used. It can take weeks or months to complete a full cap pressure curve.



AVERAGING CAPILLARY PRESSURE

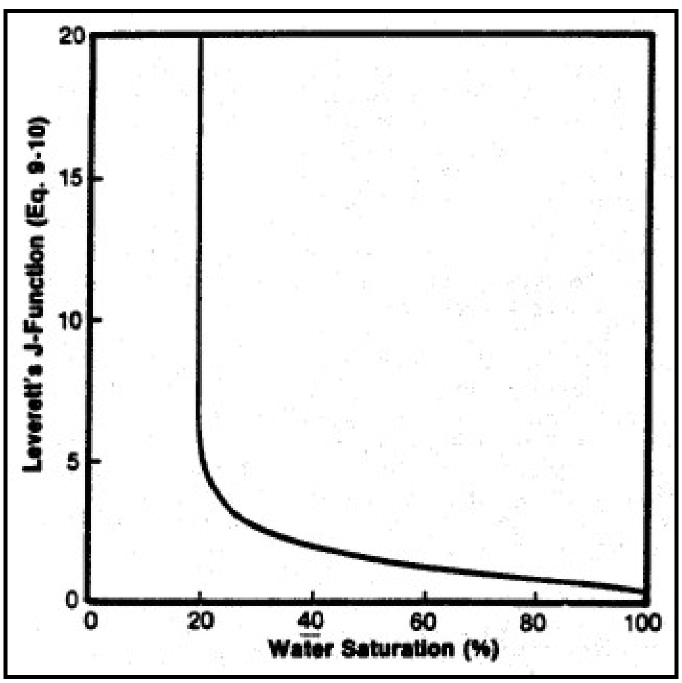
The Leverett J-function was originally an attempt to convert all capillary pressure data to a universal curve. A universal capillary pressure curve does not exist because the rock properties affecting capillary pressures in reservoir have extreme variation with lithology (rock type). But, Leverett's J-function has proven valuable for correlating capillary pressure data within a lithologic rock type.

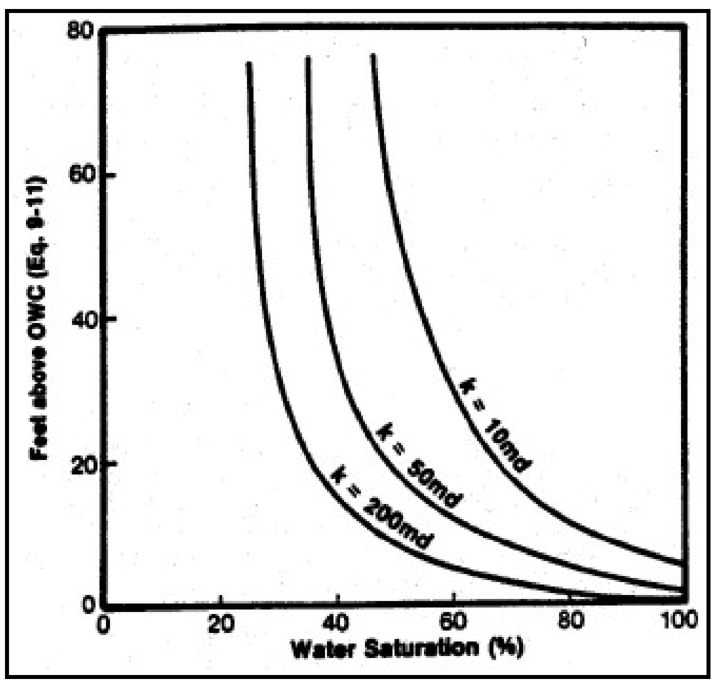
The Leverett J-Function is described by
16: Jsw = C * Pc * ((K / PHIe)^0.5) / (SIGwo * Cos (THETA))

The square root of (K / PHIe) is a function of pore throat radius. The constant C is a units conversion factor to make Jsw unitless.

By substitution and rearrangement:
17: H = Jsw * SIGwo * cos (THETA) / ((K / PHIe)^0.5) / ΔDENS

All parameters are adjusted to represent reservoir conditions. When Pc is in psi and ΔDENS is in psi/ft, then H is in feet.

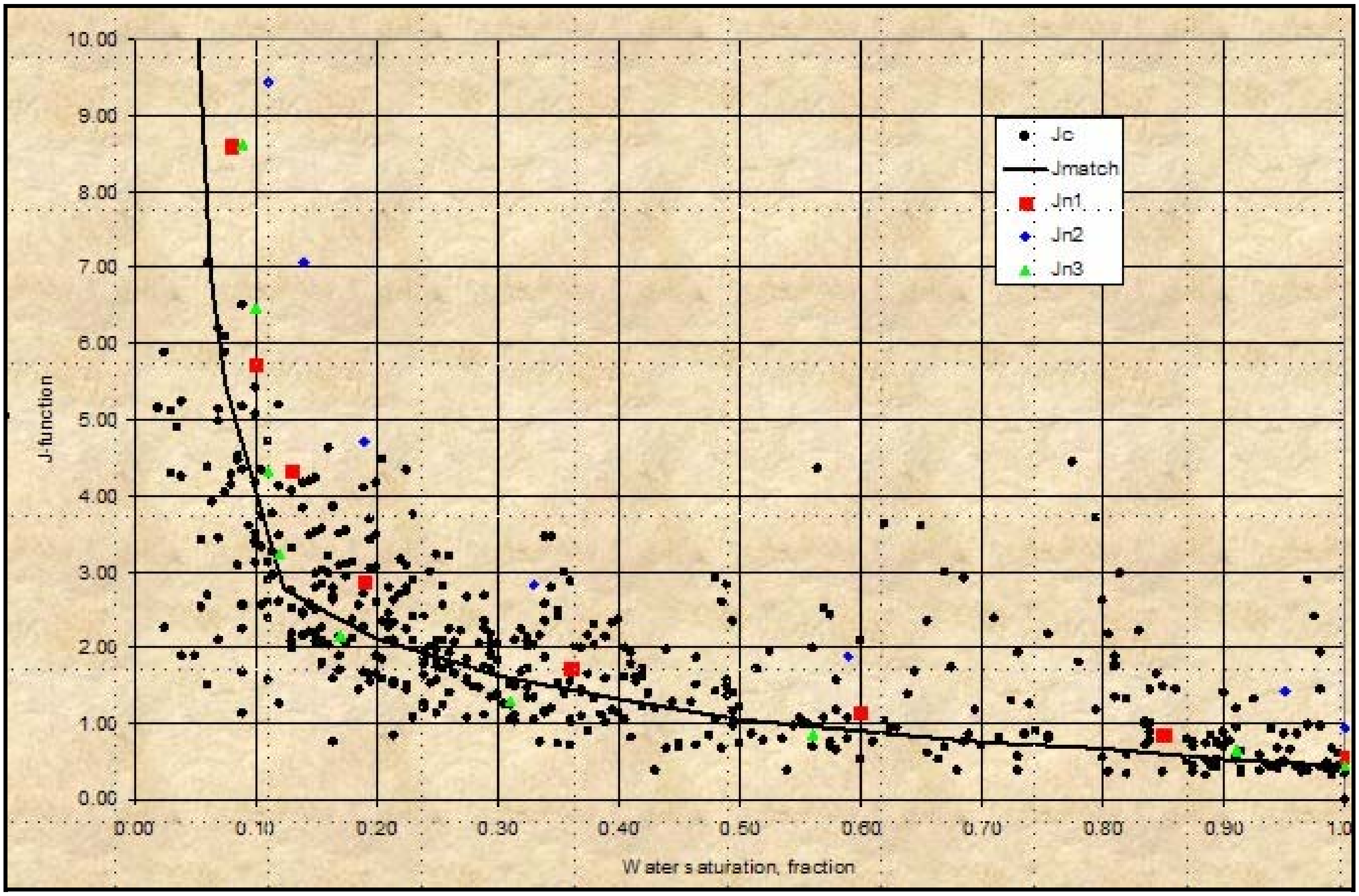




LeftL Smoothed Jsw curve versus Sw; Right: Derived saturation-height curves for various permeability values (porosity held constant) using equation 17.

J-function is useful for averaging capillary pressure data from a given rock type from a given reservoir and can sometimes be extended to different reservoirs having the same lithology. Use extreme caution in

assuming this can be done. J-function is usually not an accurate correlation for different lithologies. If J-functions are not successful in reducing the scatter in a given set of data, then this suggests that we are dealing with variation in rock type.



Leverett J-Function versus water saturation for West Texas Carbonates showing moderate spread of saturation data - rock type is not really very uniform (varying pore geometry). However, the J-function curve may be suitable for reservoir simulation purposes

CAP PRESSURE EXAMPLE - SAND AND SILT RESERVOIR

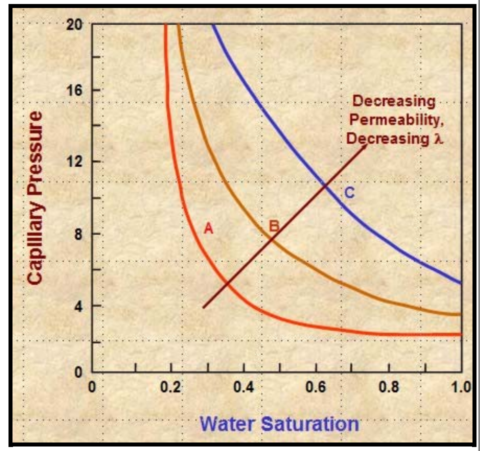
A capillary pressure (Pc) data set, along with some calculated parameters, is summarized in the table below.

CAPILLARY PRESSURE SUMMARY

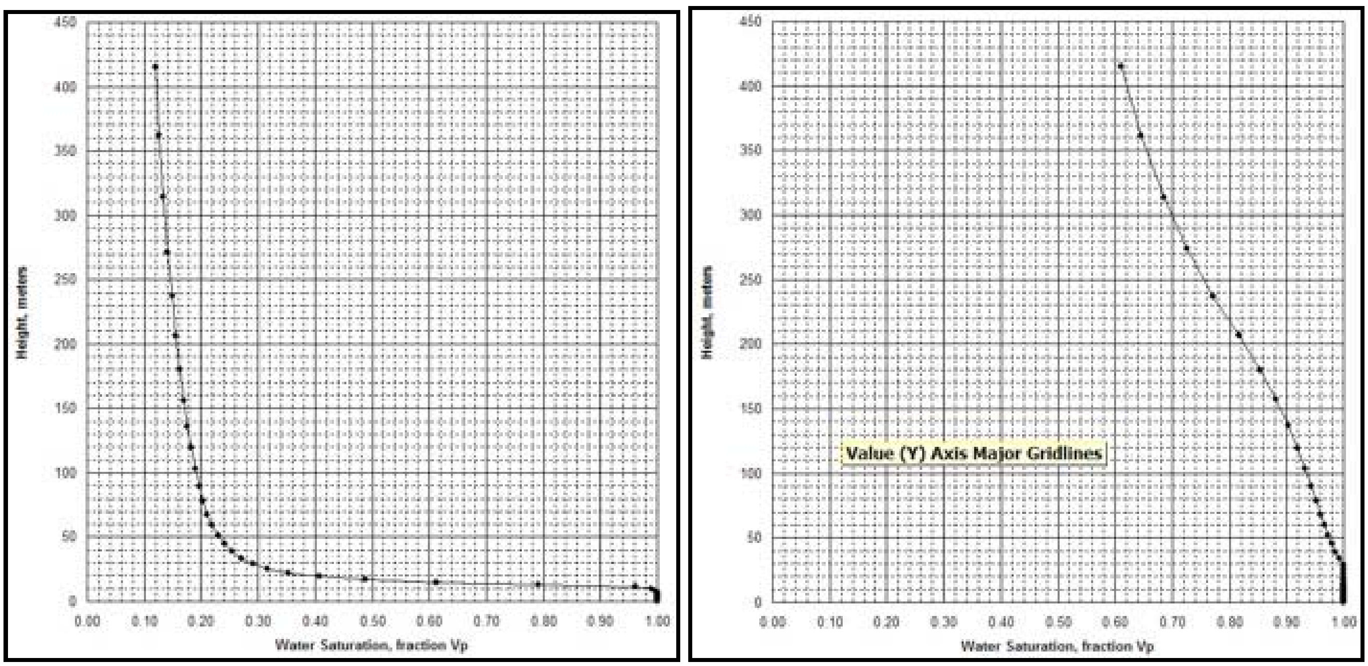
Sample	Depth	Perm	PHIe	SWir	SWir	PHI*SW	PHI*SW	sqrt/PHIe)	Pore Throat
	m	mD		425m	100m	425m	100m		Radius um
Bakken									
1	x03.5	2.40	0.118	0.12	0.19	0.014	0.022	4.51	1.358
2	x04.3	0.24	0.137	0.62	0.94	0.085	0.129	1.32	0.036
3	x04.5	0.32	0.139	0.39	0.64	0.054	0.089	1.52	0.100
4	x05.2	0.77	0.149	0.31	0.62	0.046	0.092	2.27	0.113
Average	x04.4	0.93	0.136	0.36	0.60	0.050	0.083	2.41	0.402
Torquay									

5	x16.8	0.05	0.163	1.00	1.00	0.163	0.163	0.55	0.008
6	x20.4	0.07	0.145	0.59	0.97	0.086	0.141	0.69	0.038
7	x21.8	0.09	0.174	0.79	0.96	0.137	0.167	0.72	0.019
8	x23.8	0.03	0.157	1.00	1.00	0.157	0.157	0.44	0.009
9	x31.4	0.07	0.138	0.83	0.98	0.115	0.135	0.71	0.017
Average	x24.4	0.07	0.154	0.80	0.98	0.124	0.150	0.64	0.021

In higher permeability rock, the cap pressure curve quickly reaches an asymptote and the minimum saturation usually represents the actual water saturation in an undepleted hydrocarbon reservoir above the transition zone. In tight rock, the asymptote is seldom reached, so we pick saturation values from the cap pressure curves at two heights (or equivalent) Pc values) to represent two extremes of reservoir condition.



Only sample 1 in the above table behaves close to asymptotically, as in curve A in the schematic illustration at the right. All other samples behave like curves B and C (or worse). The real cap pressure curves for samples 1 and 2 are shown below.



Examples of capillary pressure curves in good quality rock (sample 1 – left) and poorer quality rock (sample 2 – right)

The summary table shows wetting phase saturation selected by observation of the cap pressure graphs at two different heights above free water, namely 100 meters and 425 meters in this example. In this case, the 100 meter data gives water saturations that we commonly see in petrophysical analysis of well logs in hydrocarbon bearing Bakken reservoirs in Saskatchewan. This is a pragmatic way to indicate the water saturation to be expected when a Bakken reservoir is at or near irreducible water saturation. The data for the 450 meter case is considerably lower and probably does not represent reservoir conditions in this region of the Williston Basin.

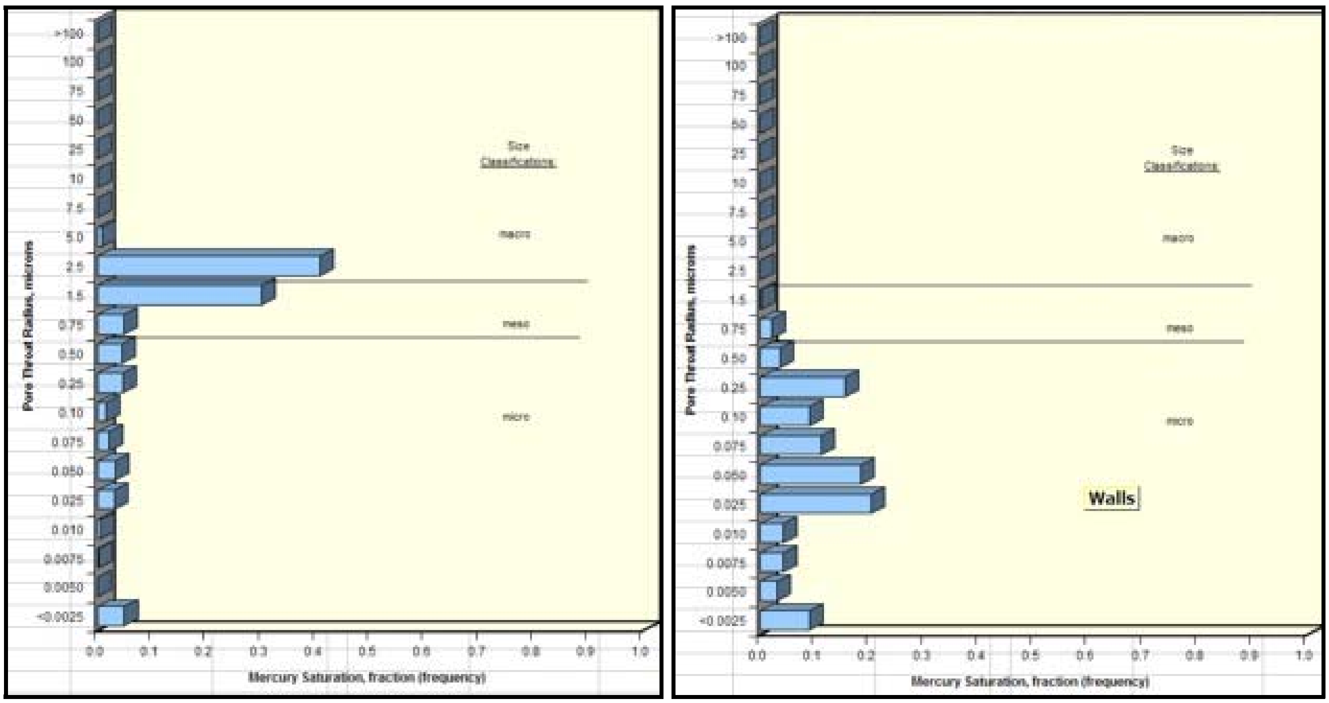
Two other columns in the table are calculated from the primary measurements.

The first is the product of porosity times saturation, PHI*SW, often called Buckle's Number. It is considered to be a measure of pore geometry or grain size. Higher values are finer grained rocks. These values vary

considerably in the Bakken, between low and medium values, indicating the laminated nature of the silt / sand reservoir. The values in the Torquay are uniformly high, indicating that the reservoir is poor quality in all samples.

The second is the square root of permeability divided by porosity, sqrt(Kmax/PHIe), which is another measure of reservoir quality, directly proportional to pore throat radius and Pc. High numbers represent good connectivity and low values show poor connectivity. Again, the Bakken shows the variations due to laminations, and the Torquay shows low values and unattractive reservoir quality.

Average pore throat radius and detailed pore throat distribution data are now routinely available in the capillary pressure spreadsheet provided by the core analysis laboratory. Examples are shown below.



Examples of pore throat radius distribution in good quality rock (sample 1 – left) and poorer quality rock (sample 2 – right)

By comparing cap pressure and pore throat distribution graphs from each sample with the quality indicator values in the summary table, it becomes more evident as to which parameters in a petrophysical analysis might be the best indicator of reservoir quality. Since both Buckle's Number and the Kmax/PHIe parameter can be determined from logs, it has been relatively common to assess reservoir quality from these parameters as a proxy for capillary pressure and pore throat measurements.

However, in thinly laminated reservoirs like the Bakken, this is not always possible since the logging tools average 1 meter of rock. This means we cannot see the internal variations of rock quality evident in the core data.

"META/SCAL" Spreadsheet -- Capillary Pressure Summary

This speardsheet is used to summarize Capillary Pressure data, with rossplot to find SWir.

DOWNLOAD Capillary Pressure Analysis Spreadsheet

META/SCAL

A Knowledge Based System For Formation Evaluation

Spectrum 2000 Mindware, Rocky Mountain House, AB, T4T 2A2 1-403-845-2527

c. E.R. Crain, P.Eng. META/SCAL HEADER DATA 15Dec2005

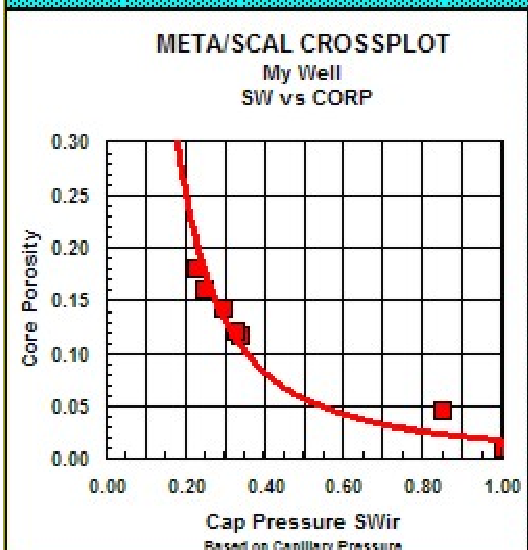
CLIENT NAME	Spectrum 2000 Mindware Ltd		ANALYST	E. R. Crain, P.Eng.	
POOL/AREA	Sample	Alberta	DATE	28-Oct-2011	
WELL/PROJECT	My Well		ZONES	MyZone1	
WELL LOCATION	01-02-03-04W5			MyZone3	
COMPUTATION	Core Summary		TESTED	OIL	
DOS FILE NAME	4METACOR	<- 8 Char Only	UNITS	M	(MorE)

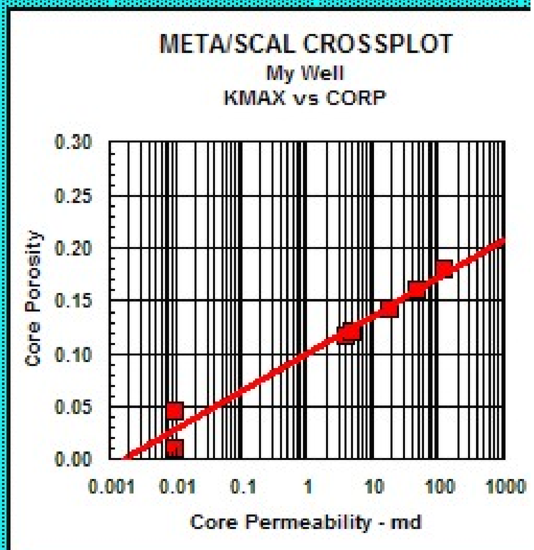
Alt>Edit>GoTo = Help META/SCAL ANALYSIS PARAMETERS

Core #	Top Depth		Bottom Depth		ecovered		Analysis #
1	2150.0	meters	2160.0	meters	7.10	meters	87465-1
2	2160.0	meters	2175.0	meters	9.00	meters	87492-3
3		meters		meters		meters	
4		meters		meters		meters	
5		meters		meters		meters	

META/SCAL RAW DATA

11-36-72-8W6						Cap Pressure		
DEPTH meters	BOTTOM meters	K-MAX md	K-90 md	K-VRT md	POR frac	SWmin frac	PHIe*Sw frac	DESCRIPTION
Enter 0.01 if Perms are zero or missing								
2050.1	2053.0	0.0	0.0	0.0	0.010	1.000	0.010	SS
2053.0	2054.2	0.0	0.0	0.0	0.047	0.850	0.040	SS
2054.2	2055.7	4.1	4.1	4.1	0.118	0.340	0.040	SS, DOL
2055.7	2056.5	18.5	18.5	18.5	0.143	0.300	0.043	SS
2056.5	2057.4	123.7	123.7	123.7	0.180	0.230	0.041	SS
2057.4	2058.0	48.7	48.7	48.7	0.160	0.250	0.040	SS
2058.0	2059.8	5.0	5.0	5.0	0.121	0.330	0.040	SS





Sample of "META/SCAL" spreadsheet for summarizing capillary pressure data.